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Linking Pupil Size Modulated by Global Luminance and Motor Preparation to Saccade Behavior

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Abstract—Saccades are rapid eye movements that are used to move the high acuity fovea in a serial manner in the exploration of the visual scene. Stimulus contrast is known to modulate saccade latency and metrics possibly via changing visual activity in the superior colliculus (SC), a midbrain structure causally involved in saccade generation. However, the quality of visual signals should also be modulated by the amount of lights projected onto the retina, which is gated by the size of the pupil. Although absolute pupil size should modulate visual signals and in turn affect saccade responses, research examining this relationship is very limited. Besides, pupil size is associated with motor preparation. However, the role of pupil dilation in saccade metrics remains unexplored. Through varying peripheral background luminance level and target visual contrast in the saccade task, we investigated the role of absolute pupil size and baseline-corrected pupil dilation in saccade latency and metrics. Higher target detection accuracy was obtained with lower background luminance level, and larger absolute pupil diameter correlated with smaller saccade amplitude and higher saccade peak velocities. More interestingly, the comparable modulation between pupil dilation and stimulus contrast was obtained, showing larger pupil dilation (or higher contrast stimuli) correlating with faster saccade latencies, larger amplitude, higher peak velocities, and smaller endpoint deviation. Together, our results demonstrated the influence of absolute pupil size induced by global luminance level and baseline-corrected pupil dilation associated with motor preparation on saccade latency and metrics, implicating the role of the SC in this behavior.

Key words: pupil light reflex, superior colliculus, pupillometry, saccade preparation, stimulus contrast, saccade latency and metrics.

INTRODUCTION

To efficiently navigate the visual world, eye movements that bring the objects of interests to the fovea, enabling high-resolution visual images of relevant stimuli for subsequent information processing, are needed. Saccades are rapid movements of the eyes that occur about 3 times per second to abruptly change the point of fixation. Saccades are modulated by stimulus salience (Itti and Koch, 2001; Borji et al., 2013), and stimulus contrast, as a primitive saliency component, is known to affect saccade responses (Ludwig et al., 2004; Marino and Munoz, 2009; Kowler, 2011). The midbrain superior colliculus (SC) is causally involved in the control of saccadic eye movements (Gandhi and Katnani, 2011), and the SC, together with other structures, is also important to saccade metrics (Sparks and Barton, 1993; Robinson and Fuchs, 2001; Sparks, 2002). Through changing SC activity via varying stimulus contrast, research has found that SC visual activity is modulated by stimulus contrast, and correlates to changes in saccade latency and kinematics (Marino et al., 2012, 2015).

Pupil size changes mainly in response to the global luminance level to regulate the amount of lights projected onto the retina to optimize the trade-off between visual sensitivity and acuity (Denton, 1956; Campbell and Gregory, 1960; Woodhouse and Campbell, 1975; Laughlin, 1992). Studies used an artificial aperture to manipulate pupil size has found that pupil size that is close to natural pupil size at a given luminance level has better performances in visual detection and discrimination (Campbell and Gregory, 1960; Woodhouse and Campbell, 1975). A study that manipulates peripheral brightness level to naturally change pupil size has further demonstrated better detection performance when pupil size is larger, whereas better discrimination performance

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when pupil size is smaller (Mathôt and Ivanov, 2019). Although these results suggest that the visual signals should be modulated by the size of the pupil and in turn affects visual detection and discrimination performances, how do visual signals modulated by absolute pupil size influence saccade behavior is yet established.

To understand the relationship between absolute pupil size and the saccade response, a recent study has manipulated the size of the pupil through changing background luminance before target appearance in the interleaved pro- and anti-saccade task (Cherng et al., 2020b). Larger saccade amplitude and higher saccade peak velocities are noted when background luminance is increased (i.e. smaller pupil diameter), compared to when it is unchanged. However, the differences in absolute pupil size across background luminance conditions are relatively small, which could eliminate some other saccade effects. Moreover, due to changes in the whole background luminance during the trial, larger arousal fluctuations should have occurred when background luminance is changed that may have confounded observed effects between absolute pupil size and saccade behavior, as arousal level can affect pupil size and saccades (Bradley et al., 2008; DiGirolamo et al., 2016).

Pupil size is also modulated by cognitive and affective processes (Loewenfeld, 1999; Bradley et al., 2008; Eckstein et al., 2017; van der Wel and van Steenbergen, 2018). Research has revealed that pupil dilation is linked to motor preparation (Richer et al., 1983; Richer and Beatty, 1985). In particular, pupil dilation is larger during anti-saccade, compared to, pro-saccade preparation, and trial-by-trial correlations between pupil dilation and saccade reaction times (SRT) have been consistently observed (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Dalmaso et al., 2020). Although these results reveal a coupling between pupil dilation and SRT (Wang and Munoz, 2015), the link between pupil dilation and saccade kinematics is yet to be examined.

The goal of this study is to understand the influence of pupil size modulated by global luminance level or associated with motor preparation on saccade latency and metrics via varying peripheral brightness level and target stimulus contrast (Fig. 1). We hypothesize that the modulation between absolute pupil size and stimulus contrast on saccade behavior should be similar, as visual signals should be enhanced by higher stimulus contrast as well as larger pupil size. Thus, larger pupil diameter induced by lower peripheral luminance level should correlate with better target detection and facilitated saccade responses. Moreover, pupil dilation associated with saccade preparation, arguably mediated by the SC (Wang and Munoz, 2015), should correlate not only to SRTs, but also to saccade metrics, because the SC also plays a role in saccade kinematics (Marino et al., 2012).

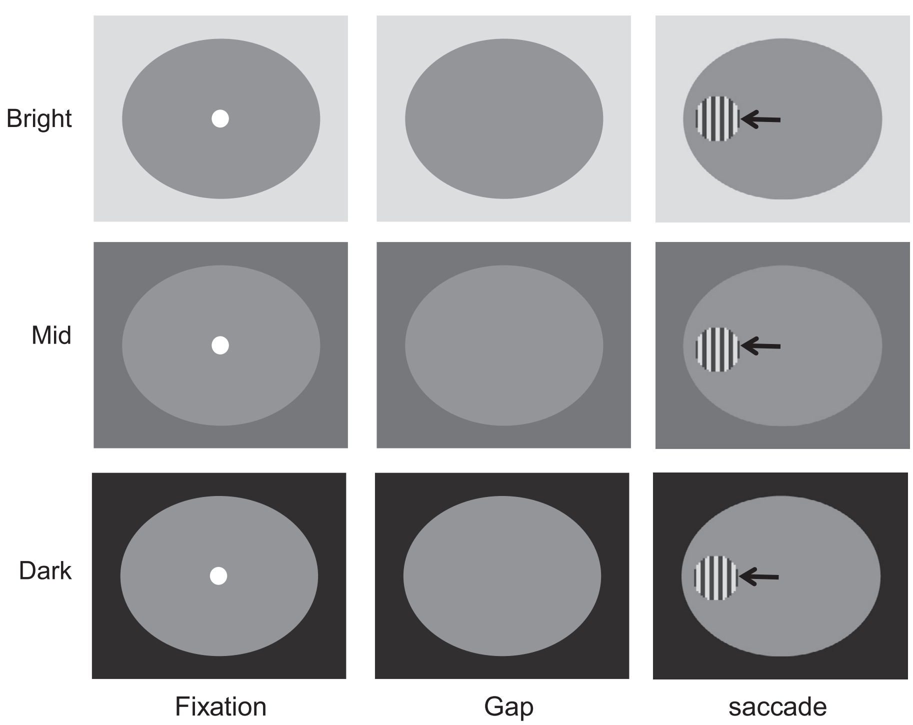

Fig. 1. Experimental paradigm. Each trial began with the appearance of a central FP on a background (Bright: 129 cd/m^2; Mid: 38 cd/m^2; Dark: 0.1 cd/m^2) with a large circle (3.5 cd/m^2). After a delay, a blank screen was presented for 200 ms (gap) before target stimulus presentation, and participants were then required to move their eyes toward the target. Note that the displayed squarewave grating circle here is only for illustration of the paradigm.

EXPERIMENTAL PROCEDURES

Experimental setup

All experimental procedures were reviewed and approved by the Institutional Review Board of the Taipei Medical University, Taiwan, and were in accordance with the Declaration of Helsinki (World Medical Association, 2001). Forty participants (mean age: 22.2, SD: 3.6 years, 23 males) were recruited via an advertisement posted on the National Central University website. Participants had normal or corrected-to-normal vision and were naïve regarding the purpose of the experiment. Participants provided informed consent and were compensated financially for their participation.

Recording and apparatus

Participants were seated in a dark room. Eye position and pupil size were measured with a video-based eye tracker (Eyelink-1000, SR Research, Osgoode, ON, Canada) at a rate of 500 Hz with binocular recording (the left eye was used for analysis because it usually had higher accuracy). Stimulus presentation and data acquisition were controlled by Eyelink Experiment Builder and Eyelink software. Stimuli were presented on an LCD monitor at a screen resolution of 1920 × 1080 pixels (60 Hz refresh rate), subtending a viewing angle of 43° × 24°, with the distance from the eyes to the monitor set at 70 cm.

Saccade task (Fig. 1)

Each trial began with the appearance of a central fixation point (FP) (0.5° diameter, a gray dot with 7.5 cd/m^2) on a background (3 luminance levels: bright: 129 cd/m^2; mid: 38 cd/m^2; dark: 0.1 cd/m^2) with a large circle at the center (21° diameter, 3.5 cd/m^2). After 1.3–1.5 s of central fixation, the FP disappeared for 200 ms (gap) before the peripheral target stimulus appeared (0.5° diameter) to the left or right of the FP (5° eccentricity on the horizontal axis). The gap period between FP disappearance and peripheral stimulus appearance was inserted, so that the size of the pupil can only influence visual signals of the target stimulus, not the FP. The insertion of a gap can also reduce fixation-related activity related to the FP (e.g., Dorris and Munoz, 1995), allowing us to examine preparatory processes related to saccade generation (Wang and Munoz, 2015). The peripheral target stimulus was a circular squarewave grating with a clear contour (6c/deg, two luminance levels), which resulted in a stimulus with same average luminance as large circular background (3.5 cd/m^2). Contrast is defined as (*Lum*high − *Lum*low)/(*Lum*high + *Lum*low), and 4 levels of stimulus contrast were used, ~12.5%, ~25%, ~50%, and ~100%. Differentiation the luminance level between the central large circle from the background allowed us to change the size of the pupil without changing the visual contrast of a target stimulus. The experiment consisted of 586 trials (10 practice trials). Target contrast condition (100%, 50%, 25%, 12.5%), peripheral background luminance condition (bright, mid, and dark), and target location (left and right) were randomly interleaved. Saccades toward either the right or left direction were combined for data analysis.

Data analysis

SRT was defined as the time from peripheral target appearance to the onset of the first saccade away from fixation defined by the moment when the eye velocity exceeded 30°/s, and with an amplitude greater than 3°. Saccade amplitude, saccade peak velocity, and endpoint saccade accuracy (angular deviation of the end position of the saccade from the correct saccadic location) of the first saccade were also analyzed. Trials were scored as correct if the first saccade after target appearance was in the correct direction (i.e., toward the target). Direction errors were identified as the first saccade away from the target location. To maintain an accurate measure of pupil size, trials with an eye position deviation of more than 2° from the central FP or with detected saccades (> 2° amplitude) from 700 ms before to target onset were excluded from analysis. When blinks were detected, pre- and post-blink pupil values were used to perform a linear interpolation to replace pupil values during the blink period (Karatekin et al., 2010; Hsu et al., 2020), and trials were discarded when two blinks occurred within a time interval of less than 500 ms to maintain the accuracy of interpolation. To remove outliers from the analysis, trials with SRTs over 1000 ms, saccade sizes over 10 deg, saccade peak velocities over 600 deg/s, or endpoint deviation over 3 deg, were considered outliers, because reaction times for visually-guided saccades are often less than 500 ms, and saccades with 5 deg amplitude are usually having ~300 deg/s peak velocities (Bahill et al., 1975; Munoz et al., 1998; Harris and Wolpert, 2006; Di Stasi et al., 2013). All above criteria resulted in the removal of ~8% of trials. To specifically examine pupil dynamics related to saccade preparation, we followed previous research that used an epoch representing the end of the pupillary responses after central fixation for baseline-correction window (Wang et al., 2016; Cherng et al., 2020a; Ayala and Heath, 2021; Ayala and Niechwiej-Szwedo, 2021; Perkins et al., 2021), because this component of pupil responses is associated with saccade preparation (Wang et al., 2015). For each trial, a baseline value was determined by averaging pupil size from 700 to 600 ms before the appearance of the target. Pupil values were subtracted from this baseline value. With this baseline-correction, we focused on the later dilation component of the pupillary responses that was more correlated to saccade preparation (Wang and Munoz, 2015). An epoch from 50 ms around the target onset (Target epoch: 0–50 ms after target appearance) was used to compare absolute pupil size (referred to as pupil diameter) and baseline-corrected pupil size (referred to as baseline-corrected pupil dilation) at target onset across four target contrast conditions in the different luminance conditions. For analysis there remained at least 5 correct trials for each condition. Thus 3 participants were excluded, leaving thirty-seven participants for final analysis.

A two-way repeated-measure ANOVA was used to examine effects of background luminance condition (bright, mid, dark) and target stimulus contrast (100%, 50%, 25%, 12.5%) on the saccade response. Bonferroni-corrected *t*-tests were used for the planned comparisons, except where indicated. A two-tailed student *t*-test was performed to compare the differences between the two conditions. Effect sizes (partial eta squared), where appropriate, are also reported. Statistical tests were performed using JASP Team (2019) and MATLAB (The MathWorks Inc., Natrick, MA, USA). Furthermore, following previous research (Watanabe et al., 2013, 2014; Cherng et al., 2020b; Chen et al., 2021), we used linear mixed models to examine the impact of absolute pupil size and baseline-corrected pupil dilation on the saccade response that allowed us to include these variables as fixed effects while taking inter-participant variability into account (Pinheiro and Bates, 2000).

RESULTS

Modulation of background luminance level on pupil diameter

Pupil size changes in response to the global luminance level, with smaller pupil sizes observed in higher levels of global luminance (Loewenfeld, 1999; McDougal and

Gamlin, 2015). Consistently, here we found that absolute pupil diameter was as a function of the global luminance level, with smaller pupil diameter in trials with higher levels of luminance (Fig. 2A). Absolute pupil diameter at the target epoch (50 ms prior to stimulus onset) was significantly influenced by background luminance level (Fig. 2B, luminance main effect: $F(2,72) = 441.771$, $p < 0.001$, $\eta_p^2 = 0.925$). Mean pupil diameter in correct trials were 3.91, 4.32, and 4.99 mm with 100% contrast stimuli, 3.87, 4.32, and 5.03 mm with 50% contrast stimuli, 3.86, 4.34, and 5.03 mm with 25% contrast stimuli, and 3.89, 4.30, and 5.01 mm with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively, showing a clear modulation of background luminance on pupil diameter across different stimulus contrast conditions (simple main effects all $p < 0.05$). Because the contrast condition should not affect pupil diameter at target onset, any effects related to target contrast are not considered for further analyses. Effects of stimulus contrast was not significant ($F(3,108) = 2.225$, $p = 0.089$, $\eta_p^2 = 0.058$), and the interaction was significant ($F(6,216) = 4.141$, $p < 0.001$, $\eta_p^2 = 0.103$).

Modulation of background luminance level on target direction rate and reaction time

If, as hypothesized, larger pupil size can increase visual sensitivity (Laughlin, 1992), then better task performance should be observed with a darker background, i.e., larger pupil diameter. As illustrated in Fig. 3A, background luminance level significantly affected direction error rates (luminance main effect: $F(2,72) = 11.686$, $p < 0.001$, $\eta_p^2 = 0.245$), with lower error rates in trials with lower background luminance levels (Bonferroni-corrected comparisons between bright and others, all $p < 0.01$). Mean error rates were 14, 13, and 9% with 100% contrast stimuli, 22, 20, and 18% with 50% contrast stimuli, 31, 32, and 29 % with 25% contrast stimuli, and 42, 34, and 38% with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. Moreover, as expected, visual contrast clearly modulated the error rate with less direction errors in trials with higher contrast stimuli (contrast main effect: $F(3,108) = 153.159$, $p < 0.001$, $\eta_p^2 = 0.810$). Interaction was also significant ($F(6,216) = 4.656$, $p < 0.001$, $\eta_p^2 = 0.115$), showing significant background luminance effects in all contrast conditions ($p < 0.05$) except for the 25% contrast condition.

Background luminance level also significantly modulated SRTs (Fig. 3B, luminance main effect: $F(2,72) = 13.603$, $p < 0.001$, $\eta_p^2 = 0.274$), with mean SRTs in correct trials being 233, 231, and 239 ms with 100% contrast stimuli, being 296, 281, and 286 ms with 50% contrast stimuli, being 333, 328, and 346 ms with 25% contrast stimuli and being 360, 356, and 384 ms with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. Interestingly, SRTs were faster in the mid background, and SRTs in the bright condition were faster than those in the dark condition (all comparisons, $p < 0.05$). As expected, visual contrast significantly affected SRTs (contrast main effect: $F(3,108) = 177.617$, $p < 0.001$, $\eta_p^2 = 0.831$), with faster SRTs for higher contrast stimuli, and interaction was also significant ($F(6,216) = 3.605$, $p = 0.008$, $\eta_p^2 = 0.091$). Together, these results suggested that, in

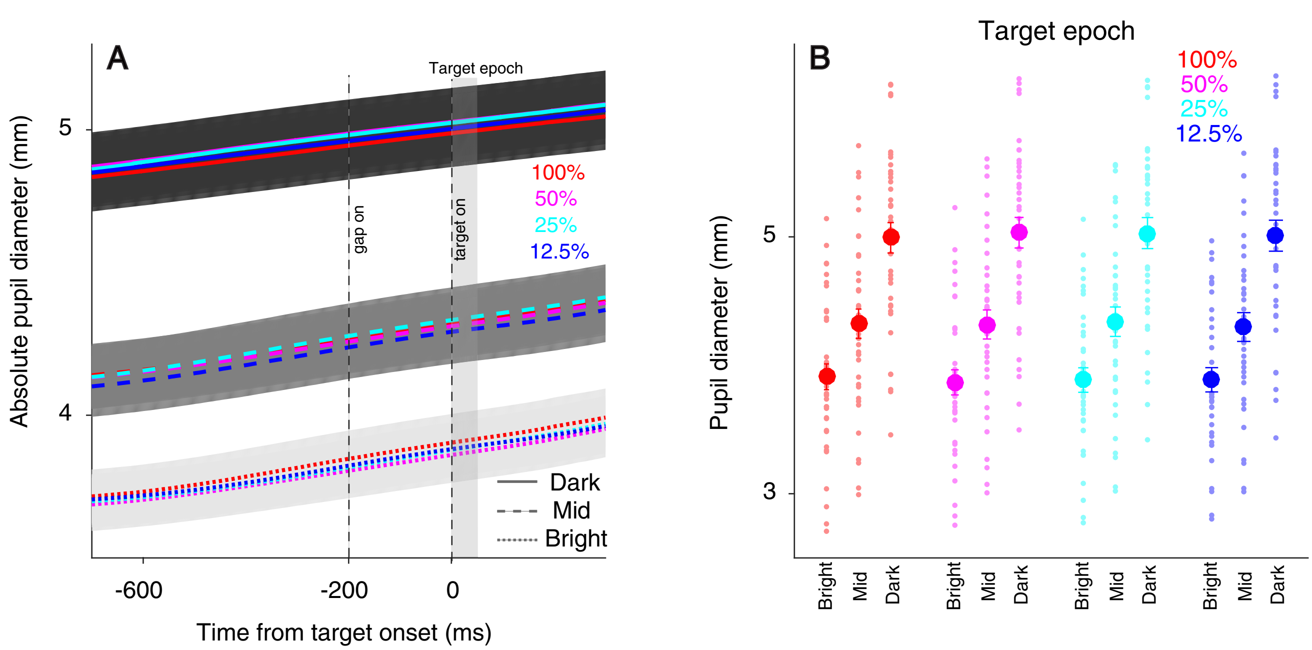

Fig. 2. Effect of background luminance on absolute pupil diameter. Absolute pupil diameter following target appearance in all conditions **(A)**. Mean absolute pupil diameter (0–50 ms after target onset) **(B)** shown for different contrast conditions with bright, mid, and dark backgrounds. In **(A)**, the shaded colored regions surrounding the pupillary response curves represent the ±standard error range (across participants) for different background luminance conditions (different contrast conditions collapsed). The target epoch for pupil size is shaded in gray. In **(B)**, the large-circle and error-bars represent the mean values ± standard error across participants. The small circles represent the mean value for each participant. 100%: 100% target visual contrast, 50%: 50% target visual contrast, 25%: 25% target visual contrast, 12.5%: 12.5% target visual contrast.

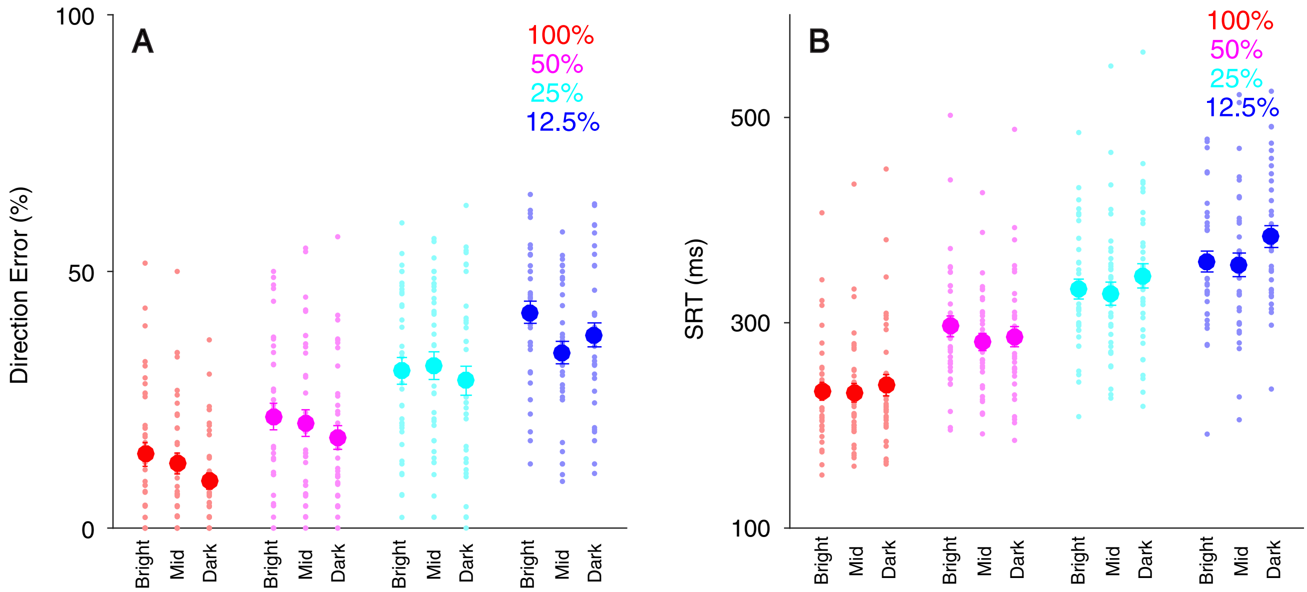

Fig. 3. Effect of background luminance and target contrast on task accuracy and SRT. Direction error **(A)** and SRT **(B)** shown for different contrast conditions with bright, mid, and dark backgrounds. The large-circle and error-bars represent the mean values ± standard error across participants. The small circles represent the mean value for each participant. SRT: saccade reaction time. 100%: 100% target visual contrast, 50%: 50% target visual contrast, 25%: 25% target visual contrast, 12.5%: 12.5% target visual contrast.

addition to the stimulus contrast effect, differences in pupil diameter also modulated target detection rate and saccade response times.

Modulation of background luminance level on saccadic metrics

To understand the influence of absolute pupil size on saccade metrics, we further analyzed saccade amplitude, peak velocity, and endpoint accuracy (deviation from the target location) in correct trials. Saccade amplitude was significantly modulated by background luminance level (Fig. 4A: luminance main effect: $F(2,72) = 40.425$, $p < 0.001$, $\eta_p^2 = 0.529$), with smaller saccade sizes with lower background luminance. Mean saccade amplitudes were 5.05, 5.00, and 4.87 deg with 100% contrast stimuli, 5.00, 4.94, and 4.82 deg with 50% contrast stimuli, 4.99, 4.90, and 4.80 deg with 25% contrast stimuli, and 4.98, 4.95, and 4.86 deg with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. Effects of target contrast and interaction were not significant (all ps > 0.13).

Background luminance also affected peak velocity (Fig. 4B: luminance main effect: $F(2,72) = 12.943$, $p < 0.001$, $\eta_p^2 = 0.264$), with higher peak velocities obtained with lower background luminance levels. Mean peak velocities were 304, 313, and 312 deg/s with 100% contrast stimuli, 304, 308, and 308 deg/s with 50% contrast stimuli, 301, 305, and 308 deg/s with 25% contrast stimuli, and 299, 307, and 307 deg/s with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. Although peak velocities were higher in the high contrast conditions, these effects were not significant (contrast main effect: $F(3,108) = 2.610$, $p = 0.074$, $\eta_p^2 = 0.068$). Other effects were negligible ($p > 0.4$).

Endpoint accuracy was significantly modulated by target contrast (Fig. 4C, contrast main effect: $F(3,108) = 3.112$, $p = 0.053$, $\eta_p^2 = 0.107$), with more accurate saccades obtained with higher contrast stimulus conditions. Mean endpoint deviation was 0.62, 0.63, and 0.69 deg with 100% contrast stimuli, 0.70, 0.70, and 0.71 deg with 50% contrast stimuli, 0.78, 0.75, and 0.76 deg with 25% contrast stimuli, and 0.79, 0.79, and 0.83 deg with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. Moreover, background luminance did not significantly modulate endpoint accuracy (luminance main effect: $F(2,72) = 2.497$, $p = 0.089$, $\eta_p^2 = 0.065$). Other effects were negligible ($p > 0.5$).

To further examine the relationship between absolute pupil diameter and saccade latency and metrics, trials across three background luminance conditions were collapsed, and trial-by-trial correlational analysis between pupil diameter at target onset and four saccade indices, namely SRT, saccade amplitude, saccade peak velocity and endpoint accuracy, in the four contrast conditions was performed. Trial-by-trial correlation coefficients for all subjects are illustrated in Fig. 5. As displayed in Fig. 5A, although some conditions showed significant correlations between pupil diameter and SRTs (parried *t*-test of *R* values against zeros), there was no reliable and consistent correlation across four target contrast conditions. In contrast, negative correlations between pupil diameter and saccade amplitude were consistently observed across all contrast conditions (Fig. 5B), suggesting that larger pupil diameter correlated with smaller saccade amplitude. Correlations between pupil diameter and saccade peak

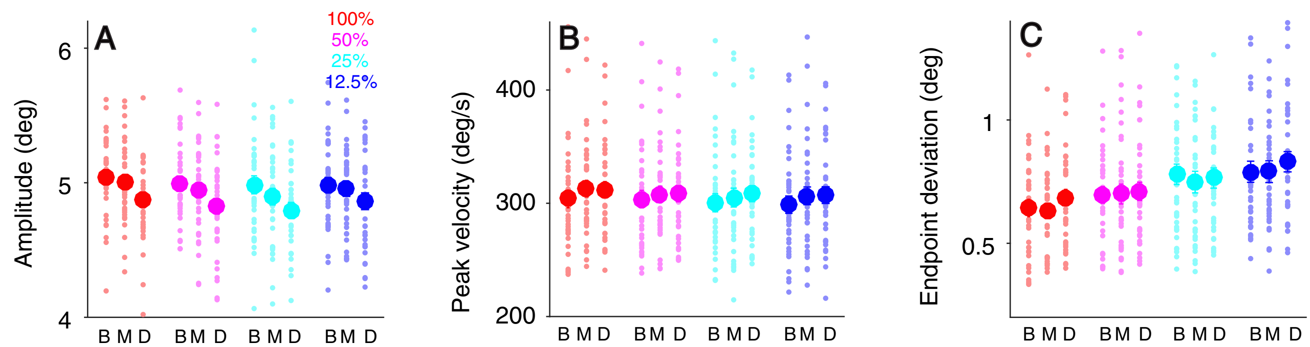

Fig. 4. Effect of background luminance and target contrast on saccade metrics. Saccade amplitude **(A)**, saccade peak velocity **(B)**, and saccade endpoint deviation **(C)** shown for different contrast conditions with bright, mid, and dark backgrounds. The large-circle and error-bars represent the mean values ± standard error across participants. The small circles represent the mean value for each participant. 100%: 100% target visual contrast, 50%: 50% target visual contrast, 25%: 25% target visual contrast, 12.5%: 12.5% target visual contrast. B: bright background, M: mid background, D: dark background.

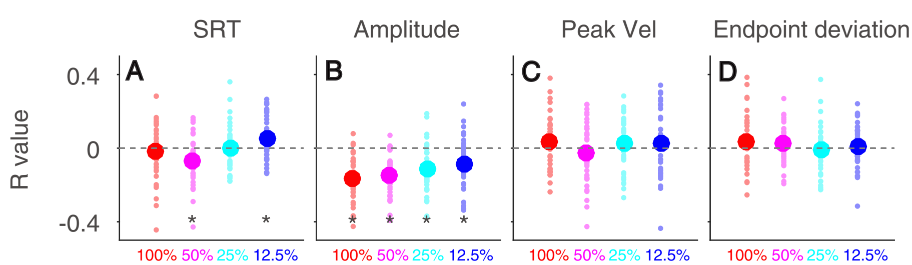

Fig. 5. Correlation between absolute pupil diameter and saccade latency and metrics. Distribution of correlation coefficients for the relationship between pupil diameter and SRT **(A)**, saccade amplitude **(B)**, peak velocity **(C)**, and endpoint deviation **(D)** for different contrast conditions (background luminance conditions were collapsed). The vertical colored large circles and error-bars represent the mean value ± standard error of correlation coefficient across all participants. The small circles represent the mean value of correlation coefficient for each participant. The horizontal dotted line represents a zero value of correlation coefficient. 100%: 100% target visual contrast, 50%: 50% target visual contrast, 25%: 25% target visual contrast, 12.5%: 12.5% target visual contrast. *indicates differences are statistically significant ($p < 0.05$).

velocity or endpoint deviation were not significant (peak velocity: Fig. 5C; endpoint deviation: Fig. 5D).

Modulation of pupil dilation related to saccade preparation on saccade latency and metrics

Pupil size is modulated by signals related to motor preparation (Richer et al., 1983; Richer and Beatty, 1985), and previous research has documented the correlation between baseline-corrected pupil dilation and SRT in a task involving pro- or anti-saccade generation (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Dalmaso et al., 2020). To understand the relationship between baseline-corrected pupil dilation and saccade latency and metrics, we further analyzed baseline-corrected pupil dilation. As illustrated in Fig. 6A, pupil dilation was observed across three background luminance conditions before target appearance, consistent with the literature (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Dalmaso et al., 2020). Mean baseline-corrected pupil dilation in correct trials at the target epoch (Fig. 6B) were 0.19, 0.18, and 0.15 mm with 100% contrast stimuli, 0.17, 0.18, and 0.15 mm with 50% contrast stimuli, 0.18, 0.20, and 0.16 mm with 25% contrast stimuli, and 0.18, 0.19, and 0.15 mm with 12.5% contrast stimuli in the bright, mid, and dark conditions, respectively. The main effect of background luminance was significant (luminance main effect: $F(2,72) = 7.211$, $p = 0.005$, $\eta_p^2 = 0.167$), and the planned comparisons revealed significant smaller pupil dilation in the dark, compared to, the mid condition ($p < 0.05$).

Similarly, trial-by-trial correlational analysis was performed (trials across three background luminance conditions were collapsed) to investigate the relationship between baseline-corrected pupil dilation and the saccade response. As illustrated in Fig. 6C–F for correlation coefficients for all subjects, the negative correlation between baseline-corrected pupil dilation and SRT was obtained across all contrast conditions (Fig. 6C), indicating that larger pupil dilation correlated with faster SRTs, which were consistent with previous studies (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Hsu et al., 2021). Moreover, positive correlations between baseline-corrected pupil dilation and saccade peak velocities were also observed across all contrast conditions (Fig. 6E), showing that larger pupil dilation correlated with higher peak velocities. Other correlations were less consistent across different contrast conditions.

Effects of absolute pupil size and baseline-corrected pupil dilation on saccade behavior using linear mixed models

To further explore the effect of absolute pupil diameter and baseline-corrected pupil dilation, we used linear mixed models that allowed us to consider data from all trials while taking inter-participant variability into account. Specifically, we investigated the contribution of absolute pupil diameter and baseline-corrected pupil

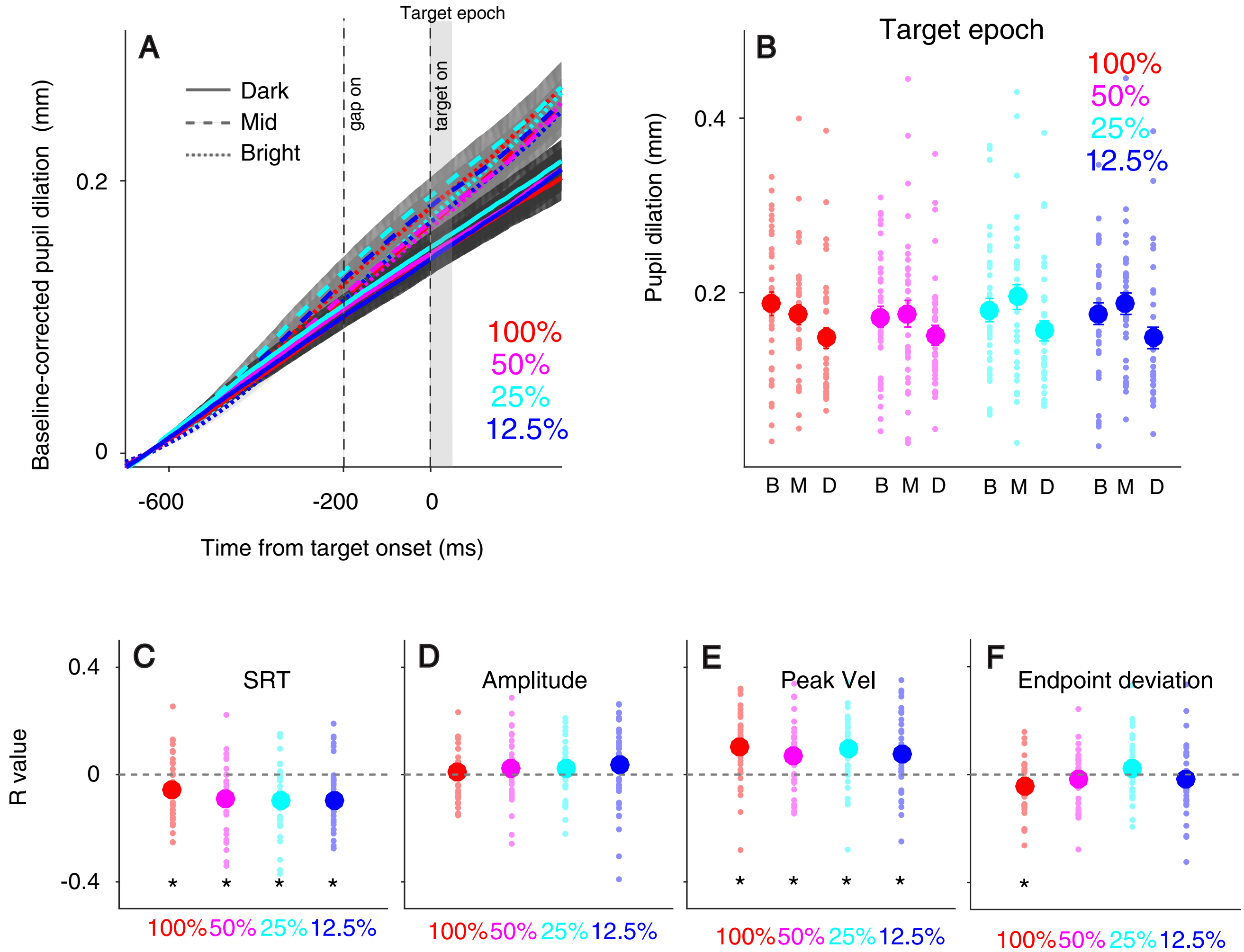

Fig. 6. Effect of baseline-corrected pupil dilation on saccade behavior. Baseline-corrected pupil size following target appearance in all conditions **(A)**. Mean pupil dilation (0–50 ms after target onset) **(B)** shown for different contrast conditions with bright, mid, and dark backgrounds. Distribution of correlation coefficients for the relationship between baseline-corrected pupil dilation and SRT **(C)**, saccade amplitude **(D)**, peak velocity **(E)**, and endpoint deviation **(F)** for different contrast conditions (background luminance conditions were collapsed). In **(A)**, the shaded colored regions surrounding the pupillary response curves represent the ± standard error range (across participants) for different background luminance conditions (different contrast conditions collapsed). The target epoch for pupil size is shaded in gray. In **(B–F)**, the large-circle and error-bars represent the mean values ± standard error across participants. The small circles represent the mean value for each participant. 100%: 100% target visual contrast, 50%: 50% target visual contrast, 25%: 25% target visual contrast, 12.5%: 12.5% target visual contrast. B: bright background, M: mid background, D: dark background. *indicates differences are statistically significant ($p < 0.05$).

dilation and target contrast on saccade behavior based on a model taking as fixed predictors target stimulus contrast [100,50,25,12.5%: 8,4,2,1], absolute pupil diameter, and baseline-corrected pupil dilation. Participants were included as a random intercept. The dependent variable, y, was one of four saccade indices, namely SRT, saccade amplitude, saccade peak velocity and endpoint deviation. Following the standard approach, four models were used from the null model to the most saturated model based upon our theoretical framework. Model comparison was performed based on AIC (Akaike information criterion) criterion. The predictors were target contrast (C), pupil diameter (P), and baseline-corrected pupil dilation (B). The linear mixed model was as follows:

$$\text{Model 1}: \; y = \beta_S$$

$$\text{Model 2}: \; y = \beta_S + \beta_1 C$$

$$\text{Model 3}: \; y = \beta_S + \beta_1 C + \beta_2 P$$

$$\text{Model 4}: \; y = \beta_S + \beta_1 C + \beta_2 P + \beta_3 B$$

where β_S is a Gaussian random variable fitted for each participant as an individual offset, and β_i are the standard coefficients of the statistical model (intercept and slopes). As shown in Table 1, Model 4 was performed better than other models, and these results were reported in details in Table 2. In SRT, $\beta_1 = -52.74$, $p < 0.001$; $\beta_2 = 0.67$, $p = 0.50$; $\beta_3 = -10.15$, $p = 3.85E{-}24$, with adjusted R-squared

Table 1. Akaike information criterion for different models

AIC	Model 1	Model 2	Model 3	Model 4
SRT	176,496	173,983	173,985	173,884
Amplitude	31,559	31,519	31,322	31,318
Peak velocity	146,949	146,903	146,890	146,794
Endpoint accuracy	19,128	18,965	18,964	18,961

Table 2. Multilevel model for saccade indices

SRT	Regression coefficient	SE	*t*	*d.f.*	*p*
'(Intercept)'	380.4064	11.0742	34.35	14,111	2.03E−248
'contrast'	−18.5086	0.3509	−52.74	14,111	0
'diameter'	1.0467	1.5532	0.67	14,111	0.5004
'dilation'	−58.0954	5.7216	−10.15	14,111	3.86E−24
Amplitude	Regression coefficient	SE	*t*	*d.f.*	*p*
'(Intercept)'	5.4783	0.0656	83.50	14,111	0
'contrast'	0.0149	0.0022	6.62	14,111	3.67E−11
'diameter'	−0.1414	0.0099	−14.24	14,111	1.12E−45
'dilation'	0.0877	0.0367	2.39	14,111	0.0168
Peak velocity	Regression coefficient	SE	*t*	*d.f.*	*p*
'(Intercept)'	289.9665	7.5598	38.36	14,111	2.81E−306
'contrast'	0.9487	0.1341	7.07	14,111	1.60E−12
'diameter'	2.082	0.5969	3.49	14,111	4.88E−04
'dilation'	21.7991	2.1886	9.96	14,111	2.71E−23
Endpoint accuracy	Regression coefficient	SE	*t*	*d.f.*	*p*
'(Intercept)'	0.756	0.0438	17.28	14,111	3.37E−66
'contrast'	−0.0187	0.0015	−12.89	14,111	8.50E−38
'diameter'	0.0117	0.0064	1.82	14,111	0.0692
'dilation'	−0.0575	0.0237	−2.43	14,111	0.0152

being 0.28. In amplitude, $\beta_1 = 6.62$, $p = 3.67\text{E}{-11}$; $\beta_2 = -14.24$, $p = 1.12\text{E}{-45}$; $\beta_3 = 2.39$, $p = 0.02$, with adjusted R-squared being 0.14. In peak velocity, $\beta_1 = 7.07$, $p = 1.60\text{E}{-12}$; $\beta_2 = 3.49$, $p = 4.88\text{E}{-04}$; $\beta_3 = 9.96$, $p = 2.71\text{E}{-23}$, with adjusted R-squared being 0.47. In endpoint deviation, $\beta_1 = -12.89$, $p = 8.50\text{E}{-38}$; $\beta_2 = 1.82$, $p = 0.0692$; $\beta_3 = -2.43$, $p = 0.0152$, with adjusted R-squared being 0.14. In addition to the pronounced contrast modulation on all saccade indices, these results revealed that absolute pupil diameter significantly modulated saccade amplitude and peak velocity, with larger pupil diameter correlating with lower saccade amplitude and higher peak velocities. Although larger pupil diameter correlated with larger endpoint deviation, this effect was not significant ($p = 0.069$). Furthermore, baseline-corrected pupil dilation, similar to target contrast, significantly correlated with all saccade indices. Specifically, larger pupil dilation correlated with faster SRTs, larger saccade amplitude, higher peak velocities, and smaller endpoint deviation.

DISCUSSION

The current study examined the effects of absolute pupil diameter induced by peripheral background luminance level and baseline-corrected pupil dilation related to motor preparation on saccades made to a visual stimulus. Absolute pupil diameter was clearly modulated by background luminance level, showing larger pupil diameter in trials with lower background luminance. In addition to the target contrast modulation on saccade behavior, differences in pupil diameter also affected saccade responses. Target detection rates were higher with lower background luminance, and SRTs were also modulated by background luminance level. Moreover, background luminance level modulated saccade kinematics, with smaller amplitude and higher peak velocities observed in trials with lower background luminance level. Besides, baseline-corrected pupil dilation significantly modulated saccade latency and metrics. Modeling results further suggested that, in addition to contrast effects, both absolute pupil diameter and baseline-corrected pupil dilation significantly influenced saccade behavior. Larger pupil diameter correlated with lower saccade amplitude and higher peak velocities, whereas larger pupil dilation correlated with faster SRTs, larger saccade amplitude, higher peak velocities, and smaller endpoint deviation. Together, our results demonstrated the pupil size modulation of bottom-up (global luminance) and top-down (motor preparation) factors on saccade latency and metrics.

Modulation of absolute pupil diameter on saccade behavior

The functional role of the pupil light reflex is thought to optimize the trade-off between sensitivity and acuity of input image for visual processing (Denton, 1956; Campbell and Gregory, 1960; Woodhouse and Campbell, 1975; Laughlin, 1992). Seminal work that uses the artificial aperture to vary pupil size has shown the optimal performance in visual detection and discrimination while the diameter of the aperture is close to the nature pupil diameter at a given background luminance level (Campbell and Gregory, 1960; Woodhouse and Campbell, 1975). A recent study that varies pupil size via changing peripheral brightness level has further confirmed this idea, showing that larger pupil sizes correlates with better visual detection performances, whereas smaller pupil sizes correlates with better visual discrimination performances (Mathôt and Ivanov, 2019). These results suggest that pupil size modulates the quality of visual signals, and in turn influences visual detection and discrimination performances.

To understand whether visual signals gated by the size of the pupil could also affect saccade generation to a target stimulus, our previous study varies pupil size at target onset via changing background luminance level before the target presentation in an interleaved pro- and anti-saccade paradigm (Cherng et al., 2020b). Results show some differences between the light condition (increase in background luminance) and the no-change condition, particularly in pro-saccade trials, that is, significantly smaller peak velocities and larger saccade amplitude when pupil diameter is smaller. However, because of the radical change in background luminance during the trial in the light or dark condition, arousal fluctuation is expectedly larger in the background-luminance change condition than that in the no change condition. Moreover, mean differences of pupil diameter across different conditions are relatively small, ~0.6 mm, and a salient target stimulus (a very bright target in the gray background) used previously could reduce the moderate effect of absolute pupil size on saccade behavior. To better investigate this question, here we varied absolute pupil diameter via changing peripheral background luminance level (Mathôt and Ivanov, 2019), such that no background changes occurred during the trial to avoid different arousal fluctuations in different background luminance conditions. With this manipulation, pupil fluctuation was relatively smaller at target onset, so we can also analyze baseline-corrected pupil dilation related to saccade preparation to investigate pupil dilation effects on saccade latency and metrics. Lastly, we systematically manipulated target contrast that allowed us to compare the effects of target contrast to those of absolute pupil diameter and baseline-corrected pupil dilation. Consistent with an idea that larger pupil sizes could increase visual sensitivity (Laughlin, 1992), we found higher detection rates when background luminance was lower. Moreover, consistent with our previous study (Cherng et al., 2020b), larger pupil diameter also correlated with smaller saccade amplitude and higher peak velocities, and these effects were significant even when the factors of stimulus contrast and baseline-corrected pupil dilation were considered in our regression model. These results suggest that absolute pupil size affected saccade targeting, likely via changing the quality of visual signals evoked by target stimuli.

Linking baseline-corrected pupil dilation to saccade latency and metrics

Pupil size is also modulated by cognitive and affective processes (Loewenfeld, 1999; Bradley et al., 2008; Eckstein et al., 2017; van der Wel and van Steenbergen, 2018). Research has found pupil dilation associated to motor preparation (Richer et al., 1983; Richer and Beatty, 1985), and subsequent studies have further demonstrated larger pupil dilation during anti-saccade, compared to, pro-saccade preparation, and the correlations between pupil dilation and SRTs are also noted (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Dalmaso et al., 2020; Hsu et al., 2021). These results together suggest that a tight coupling between pupil dilation and motor intent. Consistently, here we found that larger pupil dilation reliably correlated with faster SRTs on a trial-by-trial basis across all contrast conditions. Modeling results further suggested that pupil dilation correlated with saccade metrics, with larger saccade amplitude, higher peak velocities, and smaller endpoint deviation obtained in trials with larger pupil dilation, importantly extending the role of pupil dilation to saccade kinematics.

Neural mechanisms underlying the modulation of pupil diameter and baseline-corrected pupil dilation on saccade behavior

What neural substrates mediate this behavior? The midbrain SC is a possible candidate due to its causal role in the control of saccadic eye movements (Gandhi and Katnani, 2011). The SC receives visual signals directly from the retina and visual areas (Gamlin, 2006; Hannibal et al., 2014) and cognitive and arousal signals from the frontal eye field (FEF) and locus coeruleus (LC) as well as other areas (Edwards, 1975; White and Munoz, 2011), and projects directly to the premotor brainstem circuit to initiate the orienting response such as eye and head movements (Corneil and Munoz, 2014) and pupil dilation (Wang and Munoz, 2015). Saccade generation recruits additional brain structures such as frontal cortex and cerebellum (Gaymard et al., 1998; Robinson & Fuchs, 2001). Together with other structures such as frontal cortex and cerebellum (Gaymard et al., 1998; Robinson & Fuchs, 2001), the SC is also important to saccade kinematics (Sparks and Barton, 1993; Sparks, 2002). Through varying stimulus contrast to change the visual response of the SC, research has shown that SC visual activity is greatly modulated by stimulus contrast and correlates strongly with changes in saccade latency and metrics, with larger visual activity correlating with faster SRTs, higher peak velocities and smaller endpoint deviation (Marino et al., 2012, 2015). Given the role of the SC in saccade latency and metrics, the SC is likely underlying the observed modulation of pupil size on saccade latency and metrics. If larger pupil diameter can enhance visual detection sensitivity via increasing SC visual activities evoked by the target, then SRTs, peak velocity and endpoint accuracy should be facilitated under a lower background luminance. Although detection accuracy and saccade peak velocities increased in lower background luminance, SRTs and endpoint accuracy were not systematically modulated by background luminance level. These results suggested a differential modulation of absolute pupil size on different saccade indices. Moreover, larger pupil diameter correlated with smaller saccade amplitude, suggesting that the location of peak SC activity across the motor map was closer to the FP location in trials with larger pupil diameter. Future study with simultaneous pupil size and neuronal recordings in behaving animals is needed to directly test these predictions.

Pupil dilation is linked to saccade preparation (Jainta et al., 2011; Wang et al., 2015, 2016, 2017; Dalmaso et al., 2020; Hsu et al., 2021), arguably mediated through the oculomotor network of brain areas coordinated by the SC (Wang and Munoz, 2015). Here we found the similar modulation between target contrast and baseline-corrected pupil dilation on saccade latency and metrics. Because SC activity evoked by a visual stimulus is systematically modulated by stimulus contrast and correlates with changes in saccade latency and metrics (Marino et al., 2012, 2015), our results revealed that pupil dilation possibly correlates with SC activity. As mentioned, the SC receives extensive connections from cortical and subcortical structures including those from the FEF (White and Munoz, 2011) and LC (Edwards et al., 1979; Li et al., 2018), that, together with the SC, are importantly involved in the control of pupil size (Wang et al., 2012; Joshi et al., 2016; Lehmann and Corneil, 2016; Ebitz and Moore, 2017; Wang and Munoz, 2021). It is thus possible that the SC integrates critical control signals from these two structures to mediate the modulation of pupil dilation related to motor preparation on saccade behavior.

Pupil size is becoming a popular method for indexing neural and cognitive processes (Eckstein et al., 2017; van der Wel and van Steenbergen, 2018). Here, we demonstrated the similar modulation between baseline-corrected pupil dilation and stimulus contrast on saccade latency and metrics. In contrast, absolute pupil diameter only affected saccade amplitude and peak velocity. Given the role of the SC in saccade generation, we argue that pupil dilation related to motor preparation closely links to SC activity, but the relationship between absolute pupil size and SC activity is more complicated. Future study is needed to simultaneously record visual activity and pupil size to fully understand how does absolute pupil size modulate visual responses. As pupillometry is frequently used to infer various brain functions in health and disease, our findings, showing pupil dilation as an effective proxy of neural activity related to motor preparation, provide a potential source of behavioral biomarkers related to motor function for diseases.

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COMPETING INTEREST

The authors report no potential conflict of interest.

AUTHORS CONTRIBUTION

C.W. and C.J. designed research; C.W. and N.K. performed research; C.W. analyzed data; C.W. wrote the manuscript. All authors provided comments and edits on various drafts of the manuscript.

DATA ACCESSIBILITY

Data are available from the corresponding authors upon reasonable request.

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